

Refining Boundaries of the Segment Anything Model in Medical Images Using an Active Contour Model

Noor Nakhaei^{a,b}, Tengyue Zhang^a, Demetri Terzopoulos^a, and William Hsu^b

^aDepartment of Computer Science, University of California, Los Angeles, CA USA

^bMedical & Imaging Informatics, Department of Radiological Sciences, University of California, Los Angeles, CA USA

ABSTRACT

Accurate characterization of lesion boundaries is a critical task in diagnosing cancer. General large vision models (LVMs) such as the Segment Anything Model (SAM) have demonstrated promising performance in the semantic segmentation of natural images, but their test-time performance on medical images has been suboptimal, particularly in images with noise or poor contrast. To address this issue, we propose a test-time domain adaptation strategy that combines LVMs trained on large-scale datasets with the level-set active contours model, which can be tuned on a small subset of cases from the target dataset to yield more robust medical image segmentation performance at test-time. We showed the performance of our strategy on two datasets ISIC 2018 and CHASED_DB1, which yielded a 0.3% and 0.11% improvement in segmentation accuracy, respectively, measured by Intersection over Union. We conclude that using active contours on top of LVMs for test-time domain adaptation can improve segmentation performance.

Keywords: Image segmentation, test-time domain adaptation, large vision model, active contours model

1. INTRODUCTION

Accurately capturing a lesion's boundaries is important in cancer diagnosis. However, the contrast between the boundary and surrounding tissue is frequently low, and the similar texture of the lesion and its surrounding area complicates the task of segmenting lesions compared to delineating objects in natural images. Recent advances in medical image segmentation have yielded generalized algorithms that can delineate a variety of regions.¹ Large vision models (LVM) such as the Segment Anything Model (SAM) have been shown to perform well across different tasks.² However, these models also fail to segment the boundaries in noisy medical images or images with poor contrast, erroneously include non-relevant regions, or undersegment regions of interest. They have shown moderate to poor performance in segmenting the images different from their training sets.^{1,3,4} While the performance of segmentation models can be improved with additional supervised training examples from the target dataset, manual annotation of data is labor-intensive, and for rare cancers, few examples may be available for training. As a result of these limitations, domain adaptation techniques that require large amounts of (labeled) data can be impractical.

2. RELATED WORK

2.1 Large Vision Models

LVMs have shown robust performance in segmentation and classification tasks of natural images using datasets of natural images such as COCO and ImageNet. However, in the medical domain, fewer images are available for training a model, and even fewer are annotated. While large datasets such as MIMIC-CXR and the National Lung Screening Trial have been made publicly available, they lack annotations, such as lung and nodule segmentations. Moreover, differences in how images are acquired, even when imaging the same anatomical region, can affect the performance of segmentation methods. For example, differences in camera and lighting may cause algorithms to undersegment vessels on retinal images.

Dataset Name	Kind	Number of images	Sample Size	image size
CHASED_DB1	Retinal Vessels	28	28	999x960
DRIVE	Retinal Vessels	40	20	584x565
STARE	Retinal Vessels	20	20	700x605
ISIC 2018	Skin Lesions	2594	760	2166x3188

Table 1. Datasets

2.2 Domain Adaptation

The generalizability of machine learning algorithms, such as image segmentation, has been a longstanding problem. Wang et al.⁵ offered a boundary-aware segmentation network. However, to work across different datasets containing lesions of different sizes, they needed to re-train the model on hundreds of images in each dataset. Domain adaptation seeks to generalize a model trained on a source dataset to achieve similar performance when applied to a related target dataset.⁶

One variant of domain adaptation is test-time domain adaptation, which adapts the trained model using a subset of unlabeled cases from the target dataset.⁷ In this work, we attempt to achieve test-time domain adaptation by adding a separate component based on ACMs that require minimal parameterization using test data to improve the LVM's performance on the target dataset.

2.3 Active Contour Models

ACMs use level-set formulations that set the contour based on minimizing an energy function that fuses on the difference of the texture and/or intensity of foreground objects (e.g., lesion) and the background (e.g., surrounding tissue).⁸ Researchers have recently sought to combine ACMs and deep learning approaches. Marcos et al.⁹ combined CNNs and parametric ACMs for the segmentation of buildings in aerial images; however, their method requires manual contour initialization, fails to delineate the boundary of complex shapes precisely, and segments only single objects, all of which limit its applicability to lesion segmentation due to the irregular shapes of lesion boundaries and the potential for needing to segment multiple lesions in a single image. Hatamizadeh et al.¹⁰ implemented the Deep active lesion segmentation (DALs) framework that benefits from an improved level-set ACM formulation with a per-pixel-parameterized energy functional and a novel multiscale encoder-decoder convolutional neural networks (CNNs) that learns an initialization probability map along with parameter maps for the ACM. However, the issue with DALs network is the fixed constant in the energy function, which requires to be tuned when applied to different datasets.¹¹ presented Trainable Deep Active Contours (TDACs), an automatic image segmentation framework that unites CNNs and ACMs. The Eulerian energy function of the ACM component includes per-pixel parameter maps predicted by the backbone CNN, which also initializes the ACM. Although the entire TDAC architecture is end-to-end automatically differentiable and backpropagation trainable without user intervention, the hyperparameters are still heuristically chosen. Given these limitations, we pursued a simpler approach to integrating ACMs and a pretrained LVM, SAM, to improve its performance in medical image segmentation.

3. METHODS

In our proposed method, we demonstrate how ACMs can be combined with an existing general-purpose segmentation model trained on large datasets, transferring their weights to work on specific tasks for which only a small number of available labeled cases may exist. For each dataset, we use the pretrained model, SAM, as the initial segmentation output, then try to minimize level-set energy around the contoured boundary and have the images segmented without requiring additional supervised training for fine-tuning.

3.1 Datasets

We conduct comparison experiments on three different medical image benchmark datasets to evaluate our proposed model. We focus on these datasets for two reasons: 1) our method can be evaluated and compared to prior literature and 2) these datasets are sensitive to small differences along the boundaries.

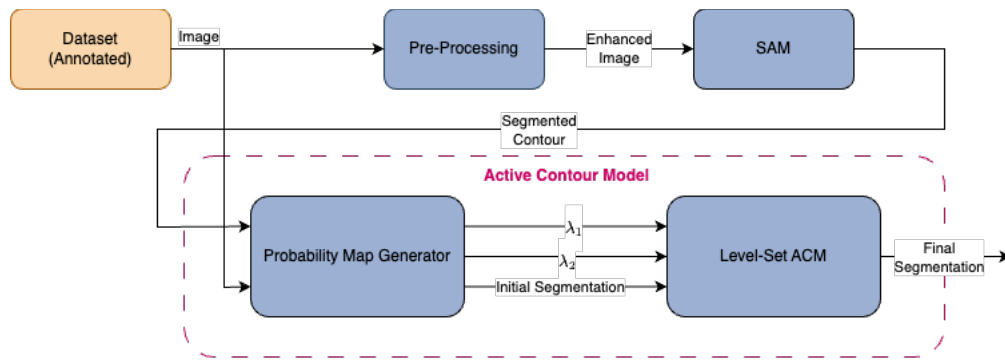


Figure 1. Segmentation Pipeline

3.1.1 Skin Lesion Segmentation

We conduct experiments involving skin lesion segmentation using the 2018 International Skin Imaging Competition (ISIC) 2018 dataset,^{12,13} This dataset consists of 2,594 RGB images of skin lesions with an average image size of 2166×3188 pixels. For our experiments, 786 cases are randomly sampled from the entire dataset.

3.1.2 Retinal Vessels

Examining retina blood vessels using fundus photography provides important information about vascular health of eye, body and brain. Numerous studies have demonstrated significant connection between the condition of the retinal blood vessels and many different diseases,¹⁴ thus making the task of retinal vessel segmentation of high importance.

The **CHASE DB1**,¹⁵ **DRIVE**,¹⁶ and **STARE**¹⁷ datasets serve as benchmarks for our approach. The CHASE DB1 dataset comprises 28 retinal images sized at 999×960 . The STARE consists of 20 retinal images with dimensions of 700×605 . The DRIVE dataset consists of 40 images sized at 584×565 , but only 20 have annotations. For this reason, we only use those 20 images to be able to report the evaluation results.

3.2 Overall Approach

We combine SAM with the level-set ACM to yield a generalizable, fully automatic medical image segmentation method to produce more accurate and detailed boundaries.

The framework includes a trained model that feeds a level-set ACM with per-pixel parameter functions.¹⁰ We used Hatamizadeh et al.'s level-set active contour model with parameter functions. Figure 1, depicts the overall pipeline. The segmented mask by the model is fed into the ACM layer, and then ACM iterations refine the contour.

3.2.1 Pre-Processing

Each image is pre-processed and smoothed using a Gaussian filter with a kernel size of 1 or 5, for retinal vessel images and skin images, respectively. A smaller smoothing kernel was chosen for the retinal vessels due to their more delicate structure.

3.2.2 Segment Anything Model

SAM was introduced by Kirillov et al.² as a large language model that was trained 11M images and 1B masks. SAM takes points or bounding boxes as a guide for focusing on each target in the image as a prompt. Figure 2, is a high-level diagram of the model. In our approach, the prompt given to the model was a bounding box that contains all of the objects in the image.

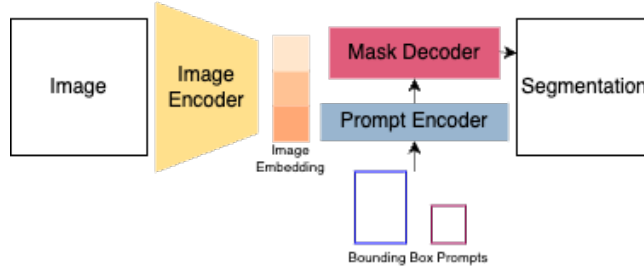


Figure 2. Proposed Pipeline

Method	IOU	Dice Score	Accuracy	Recall	Precision
SAM	0.647	0.379	0.894	0.653	0.878
SAM+ACM	0.756	0.424	0.921	0.851	0.962

Table 2. Results on the ISIC 2018 Dataset

3.2.3 Active Contour Models

The ACM level-set employed in our framework represents an extension of Chan and Vese's original method.¹⁸ The energy functional linked to a closed, time-varying contour C , denoted in $\Omega \in \mathbb{R}^2$ by the zero level set of the signed distance map $\phi(x, y, t)$, is formulated as follows:

$$E(\phi) = \int_{\Omega} \delta_{\epsilon}(\phi(x, y, t)) (\mu |\nabla \phi(x, y, t)| + \int_{\Omega} W_s F(\phi(x, y, t)) dudv) dx dy$$

Here, μ penalizes the length of C , and the energy density is defined as:

$$F(\phi) = \lambda_1(u, v)(I(u, v) - m_1(x, y))^2 H_{\Omega}^I(\phi) + \lambda_2(u, v)(I(u, v) - m_2(x, y))^2 H_{\Omega}^E(\phi)$$

To derive the curvature flow for the localized version of the uniform modeling energy, we followed the methodology outlined in:¹⁹

$$\frac{\partial \phi}{\partial t}(t) = \delta \phi(x) \int_{\Omega_y} B(x, y) \sigma \phi(y) \cdot (I(y) - u_x)^2 - (I(y) - v_x)^2 dy + \lambda \sigma \phi(x) \operatorname{div} \left(\frac{\nabla \phi(x)}{|\nabla \phi(x)|} \right)$$

In this context, x and y serve as independent spatial variables, each representing an individual point within $\Omega(x, y) = \begin{cases} 1, & ||x - y|| < r \\ 0, & \text{otherwise} \end{cases}$.

λ_1 , and λ_2 are created in the **Probability Map Generator** component by examining the segmentation outputs from the previous step. Then λ_1 , λ_2 , and the segmented mask, now as the initial contour, from the last step are passed onto the **Level-Set ACM** component.

In the **Level-Set ACM**, the contour is initialized as the segmented output mask, and through each iteration, the contour is updated to minimize the difference between the energy inside and outside of the contour.

4. RESULTS

4.1 Skin Lesion Segmentation

The initial contour was produced using SAM followed by applying the ACM for 50 iterations. As shown in Table 2, an increase in the Intersection-over-Union (IoU) is observed by 0.11 (95% CI: 0.744–0.767). The improvement in segmentation, however, is much more noticeable in cases where the SAM fails to segment the lesion in the presence of an artifacts, such as hair, or in cases the contrast between the object of interest and background is low. Figure 3 provides examples of the results of SAM+ACM on the ISIC 2018 dataset.

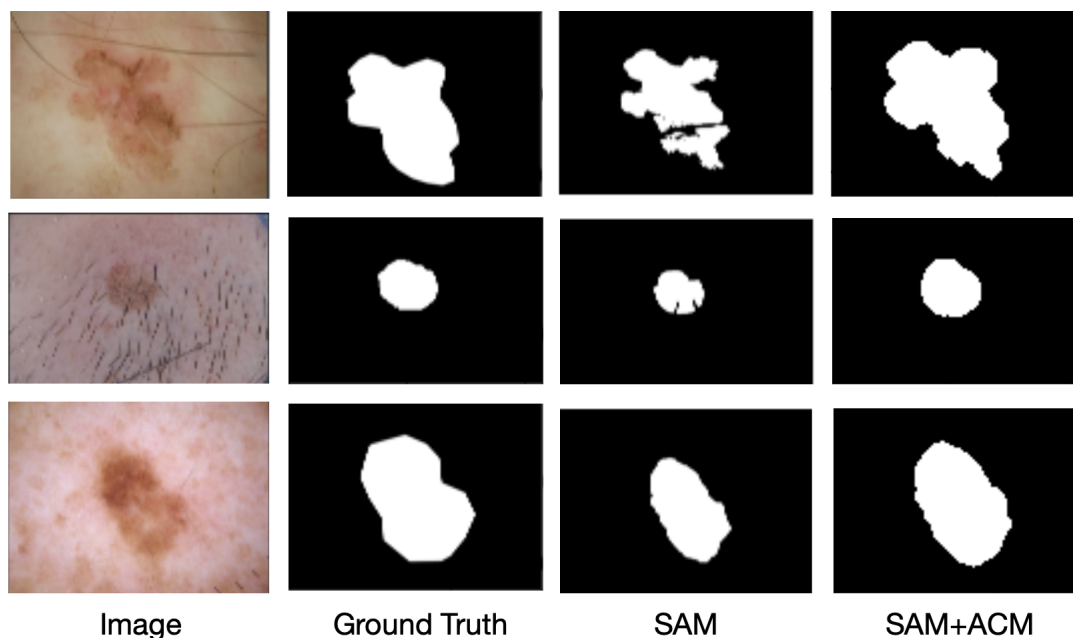


Figure 3. Examples of Improvement of Segmentation in ISIC 2018 Dataset

Dataset	Method	IOU	Dice Score	Accuracy	Recall	Precision
CHASED_DB1	SAM	0.084	0.074	0.780	0.282	0.103
CHASED_DB1	SAM+ACM	0.376	0.243	0.894	0.566	0.463
DRIVE	SAM	0.088	0.078	0.659	0.485	0.113
DRIVE	SAM+ACM	0.096	0.085	0.582	0.621	0.111
STARE	SAM	0.124	0.002	0.867	0.002	0.131
STARE	SAM+ACM	0.270	0.003	0.738	0.964	0.274

Table 3. Results on the Retinal Vessel Datasets

4.2 Retinal Vessels

Our approach was applied on three different retinal vessel datasets. First, using SAM, the initial contour was produced, and the active contour models iterated for 50 iterations. As shown in Table 3, SAM+ACM achieved the best performance on the CHASED_DB1 dataset, increasing the IoU by 0.19 (95% CI: (0.056, 0.111)). For the two other datasets, DRIVE and STARE, IoU was increased by 0.01 (95% CI: (0.069, 0.124)), and 0.15 (95% CI: (0.097, 0.151)), respectively. However, unlike the ISIC dataset, ACM fails to identify most of the vessel in cases where the SAM fails to segment any part of the vessel.

5. DISCUSSION

Our objective in this paper was to achieve domain adaptation with minimal retraining or parameter tuning, thereby enhancing the generalizability of SAM using ACMs. While we aimed to personalize object segmentation, there are certain limitations:

- We did not examine how different prompting strategies could have improved SAM's performance on medical images.
- The ACM has hyperparameters that could be tuned to a specific dataset, which we did not investigate. We also limited the number of ACM iterations to 50. For example, in the STARE and CHASED_DB datasets, failed segmentations suggest that allowing the ACM to go through more iterations may have improved results.

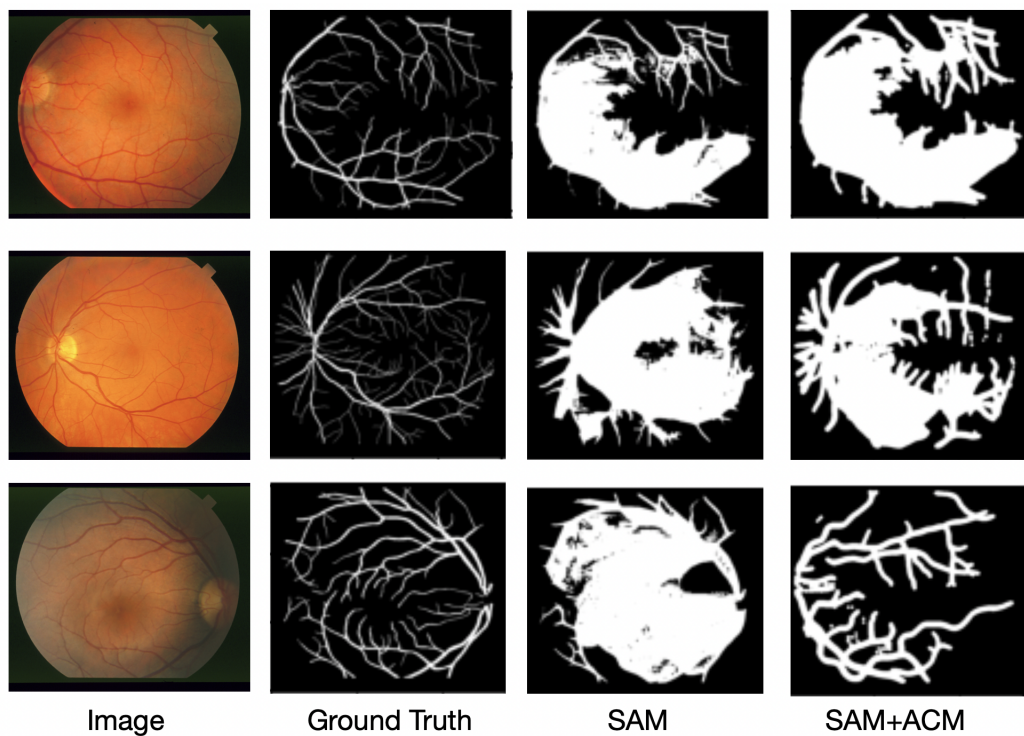


Figure 4. Examples of Improvement of Segmentation in STARE Dataset

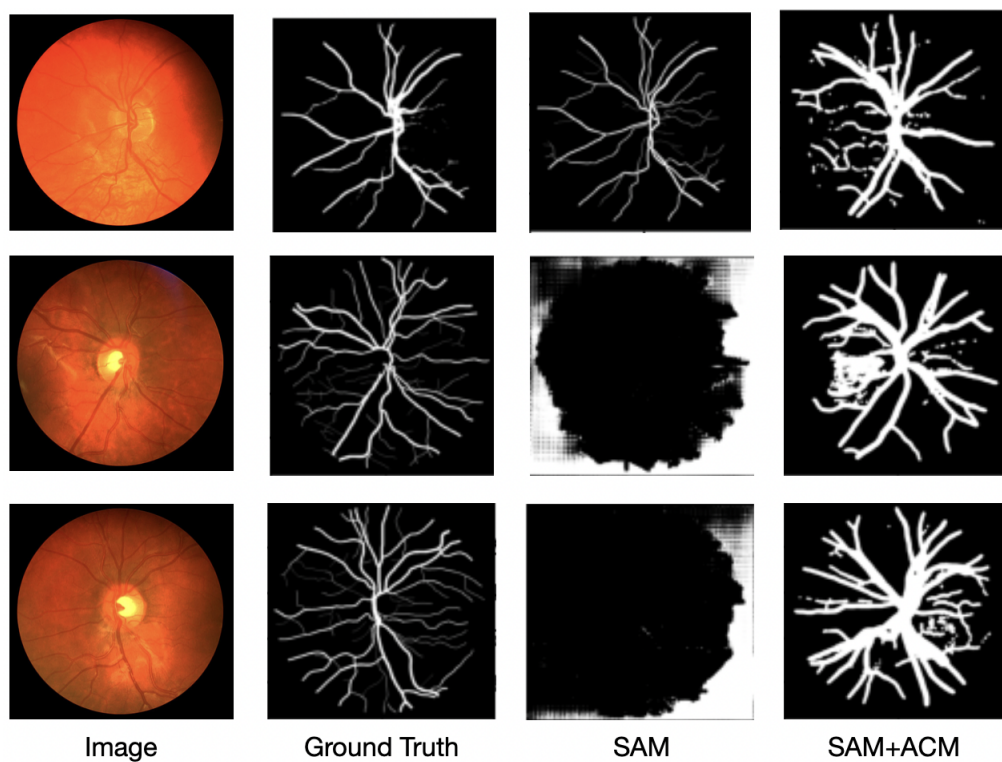


Figure 5. Examples of Improvement of Segmentation in CHASE_DB Dataset

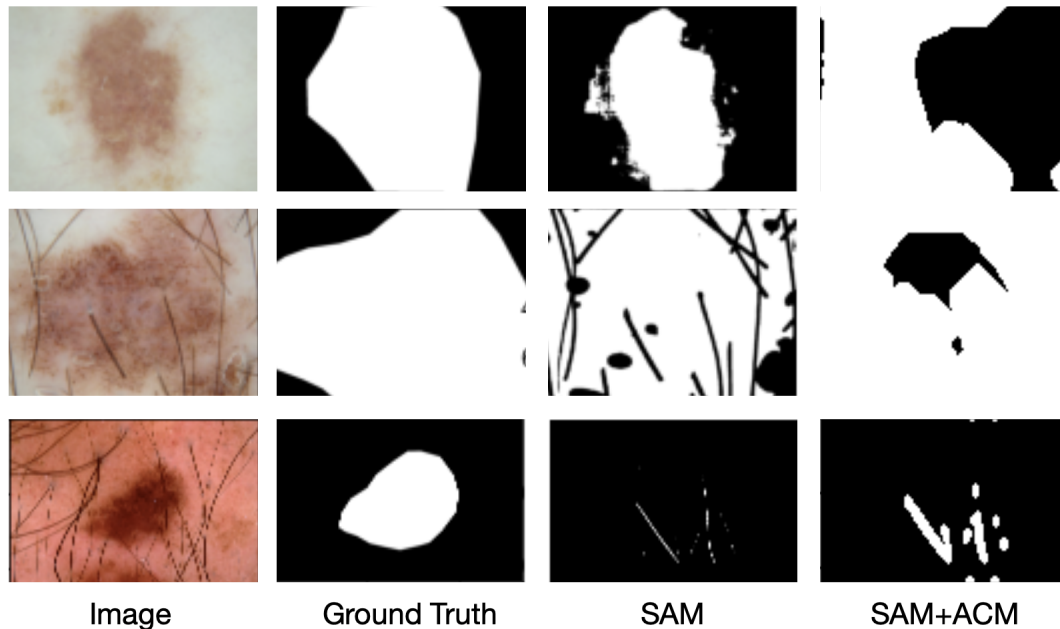


Figure 6. Examples of Cases Where Segmentation Failed in the ISIC 2018 Dataset

- Failures occurred in cases with objects that may obscure the skin lesion (e.g., hair, moles) and very narrow vessels in retinal vessels.

Figures 6, 7, 8, 9 provide examples of where the proposed method failed to segment the lesion due to the presence of hair, or when there is a low contrast between the object of interest and the background. Overall, the performance of the SAM+ACM approach on the DRIVE dataset was suboptimal. SAM failed to segment or identify vessels in all cases, and ACM proved unhelpful, as it relies on the initial contour being specified around the object.

6. CONCLUSION

Getting the precise boundary on challenging lesions when only a few labeled training cases are available is a challenging problem. Even with LVMs trained on large image datasets, fitting the models to the custom smaller datasets is necessary. Based on our experiments, we offer three major improvements: 1. Making use of the models trained on large datasets on unannotated datasets, 2. Improve segmentation results on small datasets, 3. Providing personalized segmentation while minimizing the need for additional training and hyperparameter tuning using target data.

REFERENCES

- [1] Mazurowski, M. A., Dong, H., Gu, H., Yang, J., Konz, N., and Zhang, Y., "Segment anything model for medical image analysis: an experimental study," *Medical Image Analysis*, 102918 (2023).
- [2] Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A. C., Lo, W.-Y., et al., "Segment anything," *arXiv preprint arXiv:2304.02643* (2023).
- [3] Cheng, D., Qin, Z., Jiang, Z., Zhang, S., Lao, Q., and Li, K., "Sam on medical images: A comprehensive study on three prompt modes," *arXiv preprint arXiv:2305.00035* (2023).
- [4] Shi, P., Qiu, J., Abaxi, S. M. D., Wei, H., Lo, F. P.-W., and Yuan, W., "Generalist vision foundation models for medical imaging: A case study of segment anything model on zero-shot medical segmentation," *Diagnostics* **13**(11), 1947 (2023).

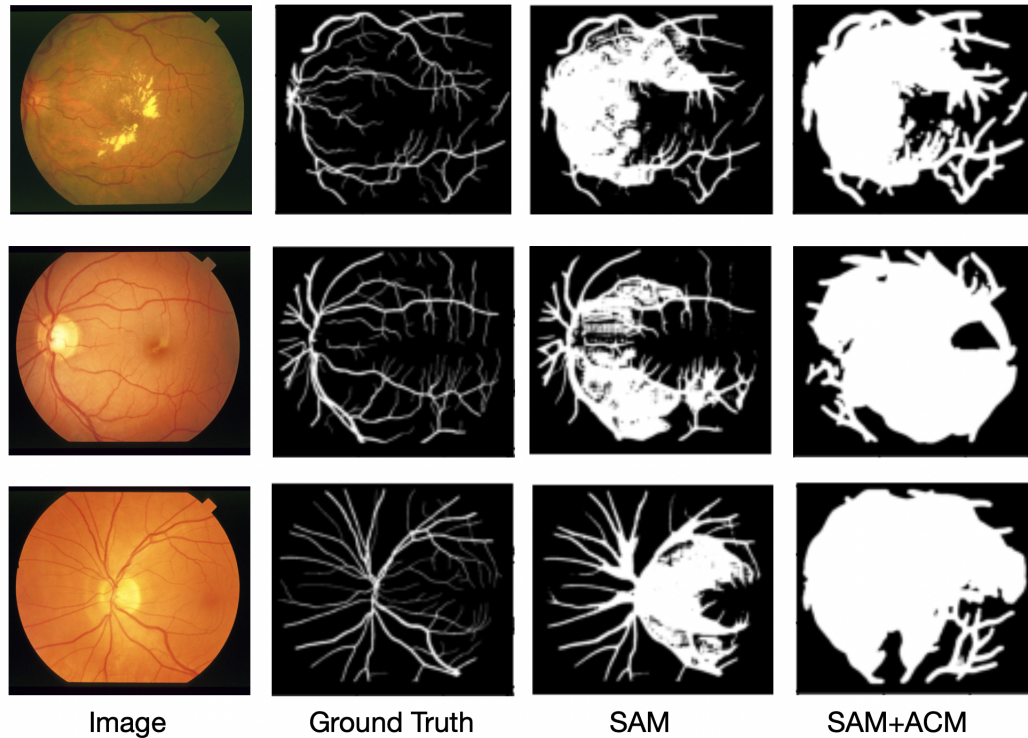


Figure 7. Examples of Cases Where Segmentation Failed in the STARE Dataset

- [5] Wang, R., Chen, S., Ji, C., Fan, J., and Li, Y., "Boundary-aware context neural network for medical image segmentation," *Medical Image Analysis* **78**, 102395 (2022).
- [6] Guan, H. and Liu, M., "Domain adaptation for medical image analysis: a survey," *IEEE Transactions on Biomedical Engineering* **69**(3), 1173–1185 (2021).
- [7] Fleuret, F. et al., "Test time adaptation through perturbation robustness," in [*NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications*], (2021).
- [8] Kass, M., Witkin, A., and Terzopoulos, D., "Snakes: Active contour models," *International journal of computer vision* **1**(4), 321–331 (1988).
- [9] Wang, Z. and Vemuri, B. C., "Tensor field segmentation using region based active contour model," in [*European conference on computer vision*], 304–315, Springer (2004).
- [10] Hatamizadeh, A., Hoogi, A., Sengupta, D., Lu, W., Wilcox, B., Rubin, D., and Terzopoulos, D., "Deep active lesion segmentation," in [*International Workshop on Machine Learning in Medical Imaging*], 98–105, Springer (2019).
- [11] Hatamizadeh, A., Sengupta, D., and Terzopoulos, D., "End-to-end trainable deep active contour models for automated image segmentation: Delineating buildings in aerial imagery," in [*European Conference on Computer Vision*], 730–746, Springer (2020).
- [12] Tschandl, P., Codella, N., Akay, B. N., Argenziano, G., Braun, R. P., Cabo, H., Gutman, D., Halpern, A., Helba, B., Hofmann-Wellenhof, R., et al., "Comparison of the accuracy of human readers versus machine-learning algorithms for pigmented skin lesion classification: an open, web-based, international, diagnostic study," *The lancet oncology* **20**(7), 938–947 (2019).
- [13] Codella, N., Rotemberg, V., Tschandl, P., Celebi, M. E., Dusza, S., Gutman, D., Helba, B., Kalloo, A., Liopyris, K., Marchetti, M., et al., "Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic)," *arXiv preprint arXiv:1902.03368* (2019).

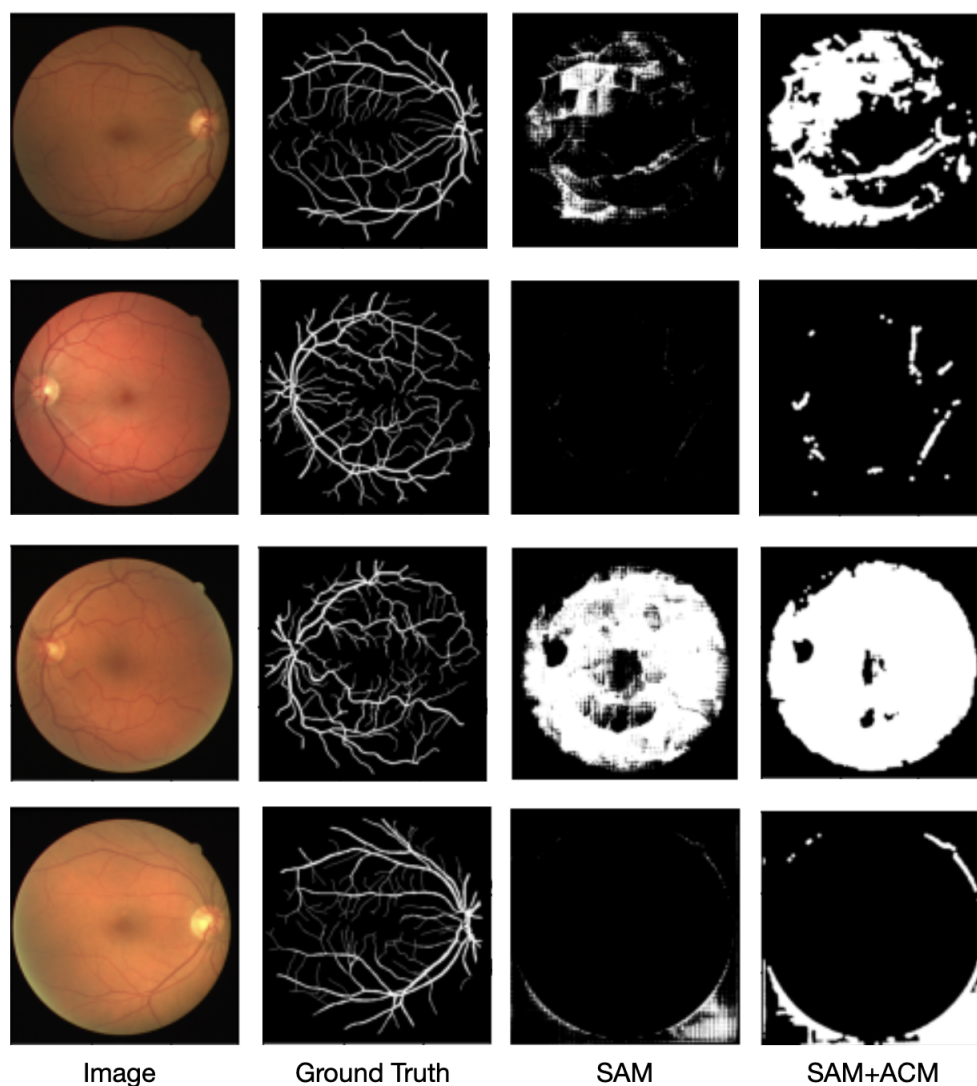


Figure 8. Examples of Cases Where Segmentation Failed in DRIVE Dataset

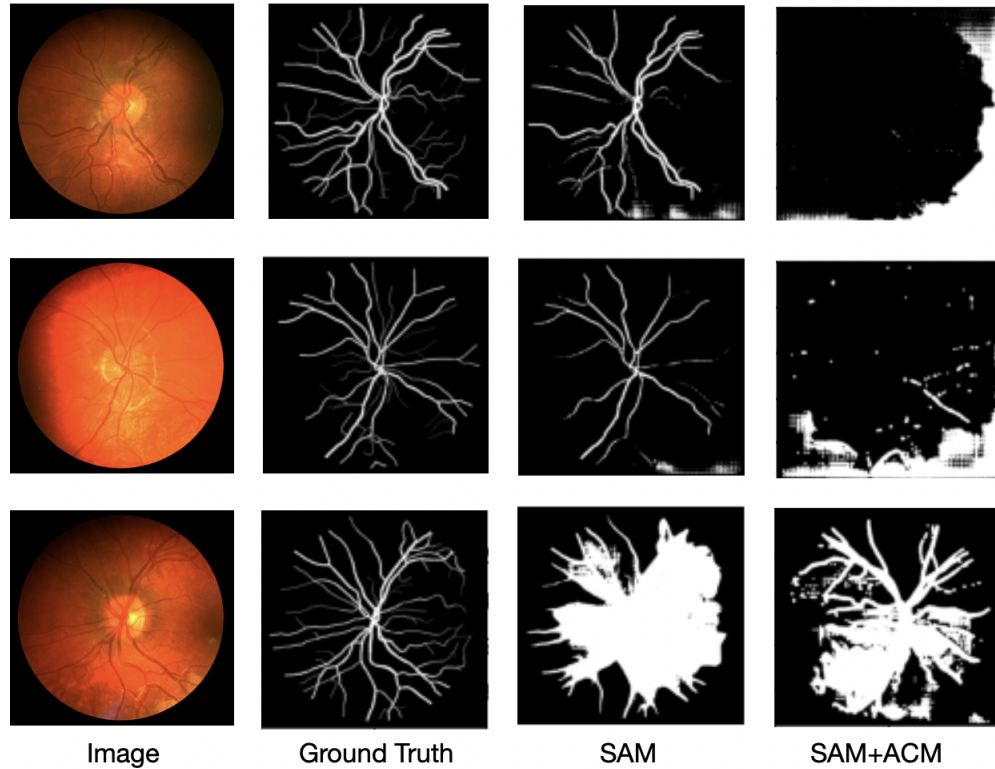


Figure 9. Examples of Cases Where Segmentation Failed in the CHASE_DB Dataset

- [14] Mookiah, M. R. K., Hogg, S., MacGillivray, T. J., Prathiba, V., Pradeepa, R., Mohan, V., Anjana, R. M., Doney, A. S., Palmer, C. N., and Trucco, E., "A review of machine learning methods for retinal blood vessel segmentation and artery/vein classification," *Medical Image Analysis* **68**, 101905 (2021).
- [15] Fraz, M. M., Remagnino, P., Hoppe, A., Uyyanonvara, B., Rudnicka, A. R., Owen, C. G., and Barman, S. A., "An ensemble classification-based approach applied to retinal blood vessel segmentation," *IEEE Transactions on Biomedical Engineering* **59**(9), 2538–2548 (2012).
- [16] Staal, J., Abramoff, M. D., Niemeijer, M., Viergever, M. A., and Van Ginneken, B., "Ridge-based vessel segmentation in color images of the retina," *IEEE transactions on medical imaging* **23**(4), 501–509 (2004).
- [17] Hoover, A., Kouznetsova, V., and Goldbaum, M., "Locating blood vessels in retinal images by piecewise threshold probing of a matched filter response," *IEEE Transactions on Medical imaging* **19**(3), 203–210 (2000).
- [18] Chan, T. F. and Vese, L. A., "Active contours without edges," *IEEE Transactions on image processing* **10**(2), 266–277 (2001).
- [19] Lankton, S. and Tannenbaum, A., "Localizing region-based active contours," *IEEE transactions on image processing* **17**(11), 2029–2039 (2008).